

Quantization (Deformation)

Introduction

Alfonso Garmendia
CRM

3

Quantization

(Deformation)⁴

1

Classical M



2

Quantum M

Alfonso Garmendia

CRM

1 Classical Mechanics

⇒

• The Model

- A Poisson manifold $(M, \{-, -\})$
- A Hamiltonian $H \in C^\infty(M)$

⇒ The evolution equation is

$$\frac{d}{dt} f = \{H, f\}$$

□

• The main example

- $M = \mathbb{R}^{2n}$ coordinates $(p_1, \dots, p_n, q_1, \dots, q_n)$

$$\{f, g\} = \sum_i \left(\frac{\partial f}{\partial p_i} \right) \left(\frac{\partial g}{\partial q_i} \right) - \left(\frac{\partial g}{\partial p_i} \right) \left(\frac{\partial f}{\partial q_i} \right)$$

- $H(p, q) = \frac{\|p\|^2}{2} + V(q)$

1 Classical Mechanics

• The Model

- A Poisson manifold $(M, \{-, -\})$

- A Hamiltonian $H \in C^\infty(M)$

⇒ The evolution equation is

$$\frac{d}{dt} f = \{H, f\}$$

• The main example

- $M = \mathbb{R}^{2n}$ coordinates $(p_1, \dots, p_n, q_1, \dots, q_n)$

$$\{f, g\} = \sum_i \left(\frac{d}{dp_i} f \right) \left(\frac{d}{dq_i} g \right) - \left(\frac{d}{dp_i} g \right) \left(\frac{d}{dq_i} f \right)$$

- $H(p, q) = \frac{\|p\|^2}{2} + V(q)$

⇒ The evolution equation is equivalent to

$$\boxed{\frac{d}{dt} p_i = \{H, p_i\} = -\frac{dH}{dq_i}}$$

$$\boxed{\frac{d}{dt} q_i = \{H, q_i\} = \frac{dH}{dp_i}}$$

} Hamilton-Jacobi equations

(Euler-Lagrange)

if these equations aren't familiar, for $H = \frac{\|p\|^2}{2} + V(q)$

⇒ $\left. \begin{array}{l} \frac{dp_i}{dt} = -\partial_{q_i} V(q) \\ \frac{dq_i}{dt} = p_i \end{array} \right\}$ Newton's Eq in $\mathbb{R}^n$

$\left[\ddot{q}_i = -\partial_{q_i} V(q) \right]$

↑
Force

↑
Acceleration



Quantum Mechanics

• The Model

- A Hilbert space L ^{states} & a subalgebra $A \subseteq B(L)$ ^{observables}
- Schrödinger operator $\hat{H}$ on L

⇒ The evolution equation is

$$i\hbar \frac{\partial f}{\partial t} = \hat{H}(f)$$

For states

$$i\hbar \frac{dQ}{dt} = [\hat{H}, Q]$$

for observables

• The main example

- $L \subset L^2(\mathbb{R}^n)$ & $A \subseteq B(L)$ (let it be vague)

$$\hat{H} = -\frac{\hbar^2}{2} \Delta + V \leftarrow V \in C^\infty(M)$$

↳ $\partial_{q_1}^2 + \dots + \partial_{q_n}^2$ the Laplacian

⇒ Evolution equation is

Quantum Mechanics

• The Model

- A Hilbert space L ^{states} & a subalgebra $A \subseteq B(L)$ ^{observables}
- Schrödinger operator $\hat{H}$ on L

⇒ The evolution equation is

$$\underbrace{i\hbar \frac{\partial}{\partial t} f = \hat{H}(f)}_{\text{For states}} \quad \underbrace{i\hbar \frac{d}{dt} Q = [\hat{H}, Q]}_{\text{for observables}} \quad \square$$

• The main example

- $L \subset L^2(\mathbb{R}^n)$ & $A \subseteq B(L)$ (let it be vague)

- $\hat{H} = -\frac{\hbar^2}{2} \Delta + V \leftarrow V \in C^\infty(M)$
 $\hookrightarrow \partial_{q_1}^2 + \dots + \partial_{q_n}^2$ the Laplacian

⇒ Evolution equation is

$$\underbrace{\text{states}}_{f \in L} \quad i\hbar \frac{\partial}{\partial t} f = -\frac{\hbar^2}{2} \Delta(f) + V \cdot f \quad \leftarrow \text{Schrödinger's equation}$$

* For observables Note that

$$\frac{d}{dt}(Qf) = (\partial_t Q)f + Q(\partial_t f) \quad \leftarrow \text{Leibniz}$$

$$\stackrel{||}{=} \frac{1}{i\hbar} \hat{H}(Qf) = (\partial_t Q)f + \frac{1}{i\hbar} Q(\hat{H}f)$$

$$\Rightarrow i\hbar (\partial_t Q)(f) = [\hat{H}, Q](f) \quad \text{eq for observables}$$



3 Example: Quantization example: Harmonic Oscillator

Quantum

for $f \in C^\infty(\mathbb{R} \times \mathbb{R})$

$$\underbrace{i\hbar \frac{d}{dt} f = -\frac{\hbar^2}{2} \frac{d^2}{dq^2} f - \frac{q^2}{2} f}_{\text{Schrödinger's}}$$

Classic

$$\underbrace{\frac{d^2}{dt^2} q = -q}_{\text{Newton's}} \quad \text{for } q \in C^\infty(\mathbb{R})$$

3 Example: Quantization example: Harmonic Oscillator

Quantum
for $f \in C^\infty(\mathbb{R} \times \mathbb{R})$

$$\underbrace{i\hbar \frac{d}{dt} f = -\frac{\hbar^2}{2} \frac{d^2}{dq^2} f - \frac{q^2}{2} f}_{\text{Schrödinger's}}$$

Classic

$$\underbrace{\frac{d^2}{dt^2} q = -q}_{\text{Newton's}} \quad \text{for } q \in C^\infty(\mathbb{R})$$

$$\text{Let } p := \frac{d}{dt} q$$

$$\Rightarrow \frac{dp}{dt} = -q$$

$$\left. \begin{array}{l} \text{Hamilton-Jacobi eq for} \\ H(p, q) = \frac{p^2}{2} - \frac{q^2}{2} \text{ in } M = \mathbb{R}^2 \end{array} \right\}$$

• This system have the symmetries

$$a(p, q) = p + q \quad \& \quad a^\dagger(p, q) = -p + q$$

$\Rightarrow$

$$\{H, a\} = p + q = a$$

$$\{H, a^\dagger\} = -p + q = -a^\dagger$$



3 Example: Quantization example: Harmonic Oscillator

Quantum
for $f \in C^\infty(\mathbb{R} \times \mathbb{R})$

$$\underbrace{i\hbar \frac{d}{dt} f = -\frac{\hbar^2}{2} \frac{d^2}{dq^2} f - \frac{q^2}{2} f}_{\text{Schrödinger's}}$$

Symmetries:

$$\Rightarrow \hat{a} = i\hbar \frac{d}{dq} + q \quad \left| \quad \hat{a}^\dagger = -i\hbar \frac{d}{dq} + q\right.$$

$$[\hat{H}, \hat{a}] = i\hbar \hat{a} \quad \left| \quad [\hat{H}, \hat{a}^\dagger] = -i\hbar \hat{a}^\dagger\right.$$

Let f_0 an eigenvector for $\hat{H} \Rightarrow \hat{H}(f_0) = \lambda f_0$

$$\Rightarrow \hat{H}(\hat{a} f_0) = (\lambda + i\hbar)(\hat{a} f_0) \quad \hat{a} f_0 \text{ \& \& } \hat{a}^\dagger f_0$$

$$\hat{H}(\hat{a}^\dagger f_0) = (\lambda - i\hbar)(\hat{a}^\dagger f_0) \quad \text{are eigen-vectors}$$

$\hat{a}$ creation $\hat{a}^\dagger$ destruction 

Classic

$$\underbrace{\frac{d^2}{dt^2} q = -q}_{\text{Newton's}} \text{ for } q \in C^\infty(\mathbb{R})$$

$$\text{Let } p := \frac{d}{dt} q$$

$$\Rightarrow \frac{dp}{dt} = -q$$

$$\left. \begin{array}{l} \text{Hamilton-Jacobi eq for} \\ H(p, q) = \frac{p^2}{2} - \frac{q^2}{2} \text{ in } M = \mathbb{R}^2 \end{array} \right\}$$

• This system have the symmetries

$$a(p, q) = p + q \quad \& \quad a^\dagger(p, q) = -p + q$$

$\Rightarrow$

$$\{H, a\} = p + q = a$$

$$\{H, a^\dagger\} = -p + q = -a^\dagger$$

3 Quantization Dream ★

Poisson Mfd $M \leftarrow \begin{matrix} Q \\ U \\ A \\ N \\ T \end{matrix} \rightarrow$ Hilbert space L_M
any function $f \in C^\infty(M) \leftarrow \begin{matrix} Q \\ U \\ A \\ N \\ T \end{matrix} \rightarrow$ operator $\hat{f}: L_M \rightarrow L_M$

such that:

(1) $f \rightarrow \hat{f}$ is $\mathbb{C}$ -linear

$$(2) [\hat{f}, \hat{g}] = i\hbar \widehat{\{f, g\}}$$

$$(3) \widehat{\frac{\|p\|^2}{2} + V(q)} = -\frac{\hbar^2}{2} \Delta + V$$

(4) for P a power series in $\mathbb{R}$

$$\widehat{P(f)} = P(\hat{f})$$

Why we want this? (1) convenience

(2) if $\partial_t f = \{H, f\}$
 $\Rightarrow \partial_t \hat{f} = \frac{1}{i\hbar} [\hat{H}, \hat{f}]$ } solutions to solutions
symmetries to symmetries

(3) It fits in the known examples

(4) It is very convenient. For example to solve

$$\partial_t f = \hat{g} f$$

the natural solution is: $f = e^{t\hat{g}} f_0$ if exists
and it would because $e^{t\hat{g}} = \widehat{e^{tg}}$

3 Quantization Dream ★

Poisson Mfd $M \leftarrow \begin{matrix} Q \\ U \\ A \\ N \\ T \end{matrix} \rightarrow$ Hilbert space L_M
any function $f \in C^\infty(M) \leftarrow \begin{matrix} Q \\ U \\ A \\ N \\ T \end{matrix} \rightarrow$ operator $\hat{f}: L_M \rightarrow L_M$

such that

① $f \rightarrow \hat{f}$ is $\mathbb{C}$ -linear

② $[\hat{f}, \hat{g}] = i\hbar \widehat{\{f, g\}}$

③ $\widehat{\frac{\|p\|^2}{2} + V(q)} = -\frac{\hbar^2}{2} \Delta + V$

④ for P a power series in $\mathbb{R}$

$$\widehat{P(f)} = P(\hat{f})$$

Why we want this? ① convenience

② if $\partial_t f = \{H, f\}$
 $\Rightarrow \partial_t \hat{f} = \frac{1}{i\hbar} [\hat{H}, \hat{f}]$ } solutions to solutions
symmetries to symmetries

③ It fits in the known examples

④ It is very convenient. For example to solve

$$\partial_t f = \hat{g} f$$

the natural solution is: $f = e^{t\hat{g}} f_0$ if exists
and it would because $e^{t\hat{g}} = \widehat{e^{tg}}$

★

PROBLEM! This Dream is a lie!

3 Real Quantization (still using $\mathbb{C}$)

$$\begin{array}{ccc} \text{Poisson Mfd } M & \leftarrow \begin{array}{|c|} \hline Q \\ \hline V \\ \hline \Delta \\ \hline N \\ \hline T \\ \hline \end{array} & \rightarrow \text{A Hilbert space } L_M \\ \forall f \in A \subseteq C^\infty(M) & \leftarrow & \rightarrow \text{an operator} \\ \uparrow & & \hat{f} : \text{Dom}(\hat{f}) \rightarrow L_M \\ \text{subalgebra} & & \cap \\ & & L_M \end{array}$$

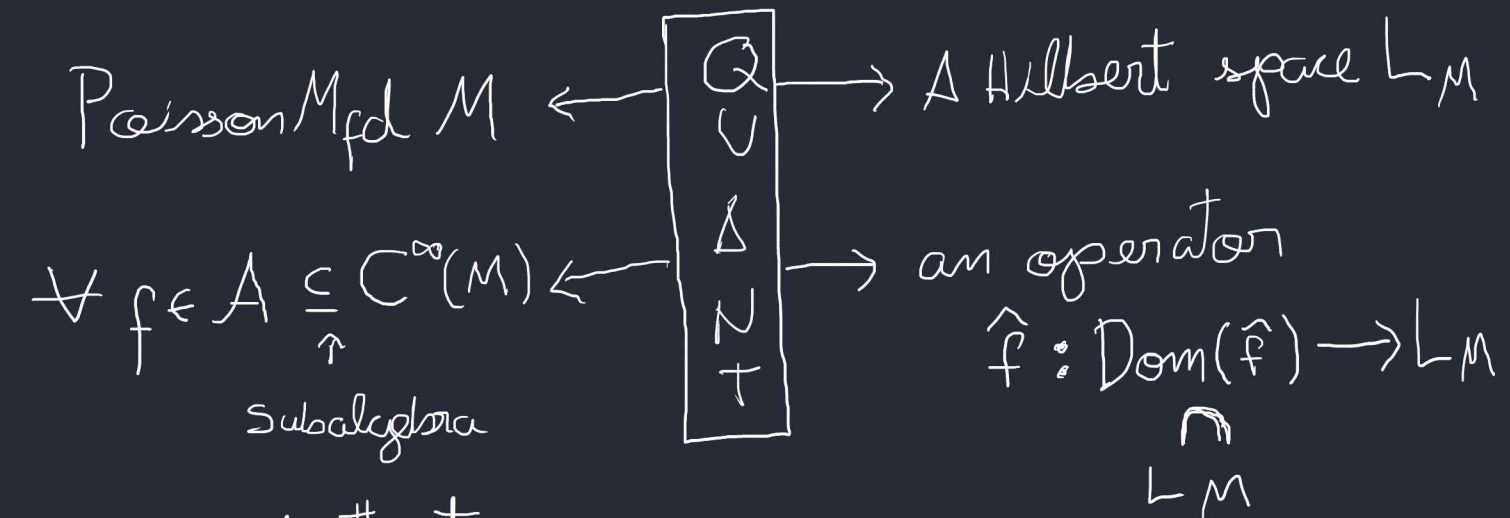
such that

- ① $f \mapsto \hat{f}$ is $\mathbb{C}$ -linear
- ② $[\hat{f}, \hat{g}] = i\hbar \widehat{\{f, g\}} + \mathcal{O}(\hbar^2)$
- ③ $\widehat{\left\{ \frac{1}{2} \|P\|^2 + V(q) \right\}} = -\frac{\hbar^2}{2} \Delta + V$
- ④ for P a power series in $\hbar$

$$P(\hat{f}) = \widehat{P(f)} + \mathcal{O}(\hbar)$$



3 Real Quantization (still using $\mathbb{C}$)



such that

- ① $f \mapsto \hat{f}$ is $\mathbb{C}$ -linear
- ② $[\hat{f}, \hat{g}] = i\hbar \widehat{\{f, g\}} + \mathcal{O}(\hbar^2)$
- ③ $\widehat{\left\{ \frac{\|P\|^2}{2} + V(q) \right\}} = -\frac{\hbar^2}{2} \Delta + V$
- ④ for P a power series in $\hbar$
 $P(\hat{f}) = \widehat{P(f)} + \mathcal{O}(\hbar)$

Example $M = \mathbb{R}^{2n}$

in $\mathbb{R}^n$ there is the Fourier transform
 for any $f \in C_c^\infty(\mathbb{R}^n)$ it is

$$\mathcal{F}(f)(p) = \int_{\mathbb{R}^n} e^{-\frac{i}{\hbar} \langle p, q' \rangle} f(q') dq'$$

- it satisfy: $\mathcal{F}(\partial_{q_i} f)(p) = \frac{-i}{\hbar} p_i \mathcal{F}(f)(p)$
- Therefore: $\mathcal{F}\left(-\frac{\hbar^2}{2} \Delta(f)\right) = \frac{\|p\|^2}{2} \mathcal{F}(f)(p)$
- This help us rewrite the Schrödinger operator as:

$$\hat{H}(f)(q) = \mathcal{F}^{-1}\left(\mathcal{F}\left(-\frac{\hbar^2}{2} \Delta(f) + V f\right)\right)_q$$

$$= \iint e^{i \text{Bl}_a(q-q')} \underbrace{H(p, q')}_{\text{orange}} f(q') dq' dp$$

4 Formal Deformation Quantization

Idea

Poisson $M \xrightarrow{DQ} C^*-\text{Algebra } A_M$
 $f \in C^\infty(M) \mapsto a_f \in A_M$

Not DQ

Gelfand-Naimark-Siegel

Rep theory

Hilbert L_M

$\mapsto \hat{f}_M: \text{Dom}(\hat{f}) \rightarrow L_M$

How

- For $M \xrightarrow{DQ} (C^\infty(M)[[\hbar]], *)$
Power series | Not usual product
- $f \in C^\infty(M) \mapsto \hat{f} = f + O(\hbar) + O(\hbar^2) + \dots$
- $f * g = f \cdot g + \hbar B_1(f, g) + \hbar^2 B_2(f, g) + \dots \quad \textcircled{1}$

such that:

- The product $*$ is associative (otherwise not a C^* -Algebra)

$$[a_f, a_g] = f * g - g * f = i\hbar \{f, g\} + O(\hbar^2)$$

This gives some restrictions

4 Formal Deformation Quantization

Idea

Poisson $M \xrightarrow{DQ} C^*-\text{Algebra } A_M$
 $f \in C^\infty(M) \mapsto a_f \in A_M$

Not DQ

Gelfand-Naimark-Segal
Rep theory $\xrightarrow{\quad} \text{Hilbert } L_M$

$\longmapsto \hat{f}_M: \text{Dom}(\hat{f}) \rightarrow L_M$

How

• For $M \xrightarrow{DQ} (C^\infty(M)[[\hbar]], *)$

Power series | Not usual product

• $f \in C^\infty(M) \mapsto \hat{f} = f + O(\hbar) + O(\hbar^2) + \dots$

• $f * g = f \cdot g + \hbar B_1(f, g) + \hbar^2 B_2(f, g) + \dots = O(\hbar^3)$ ①

such that:

• The product $*$ is associative (otherwise not a C^* -Algebra)

• $[a_f, a_g] = f * g - g * f = i\hbar \{f, g\} + O(\hbar^2)$

This gives some restrictions

• There is a cohomology theory the Hochschild Cohomology controlling the B_i in formula ① such that $*$ is associative

• $B_1(f, g) - B_1(g, f) = i\{f, g\}$

• Theorems of Kontsevich-Tamarkin-Weil-Fedorov etc...

There exist a product $*$ for any Poisson M
 $*$ is unique up to something non trivial

4 Formal Deformation Quantization

Idea

$$\begin{array}{l} \text{Poisson } M \xrightarrow{\text{DQ}} C^*-\text{Algebra } A_M \\ f \in C^\infty(M) \mapsto a_f \in A_M \end{array} \quad \left\{ \begin{array}{l} \text{Not DQ} \\ \text{Gelfand-Naimark-Segal} \\ \text{Rep theory} \quad \text{Hilbert } L_M \\ \longmapsto \hat{f}_M: \text{Dom}(\hat{f}) \rightarrow L_M \end{array} \right.$$

How

- For $M \xrightarrow{\text{DQ}} (C^\infty(M)[[\hbar]], *)$
Power series | Not usual product
- $f \in C^\infty(M) \mapsto \hat{f} = f + O(\hbar) + O(\hbar^2) + \dots$
- $f * g = f \cdot g + \hbar B_1(f, g) + \hbar^2 B_2(f, g) + \dots = O(\hbar^3)$ ①

such that:

- The product $*$ is associative (otherwise not a C^* -Algebra)
- $[a_f, a_g] = f * g - g * f = i\hbar \{f, g\} + O(\hbar^2)$

This gives some restrictions

- There is a cohomology theory the Hochschild Cohomology controlling the B_i in formula ① such that $*$ is associative
- $B_1(f, g) - B_1(g, f) = i\{f, g\}$
- Theorems of Kontsevich-Tamarkin-Weil-Fedorov etc...
- There exist a product $*$ for any Poisson M
 $*$ is unique up to something non trivial
- There is no explicit construction of $*$ for any M yet.